
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Out-of-distribution (OOD) detection is a foundational requirement for the robust-
2 ness, safety, and reliability of machine learning systems across the broad range of
3 open-world environments in which they are deployed. A growing body of work has
4 explored the use of unlabeled data to strengthen OOD detection, and this direction
5 shows considerable promise across diverse settings. Yet effectively exploiting
6 unlabeled in-the-wild data remains a general challenge, because such data arrives
7 as an unknown mixture of in-distribution (InD) and OOD samples. The absence of
8 a cleanly separated set of OOD examples makes training an optimal OOD classifier
9 difficult in essentially any realistic deployment. In this work, we introduce Medix,
10 a general framework for identifying potential outliers from unlabeled data through
11 the median operation. We build the method around the median because it delivers
12 a stable estimate of central tendency and is broadly robust to noise and outliers,
13 making it a natural and widely applicable OOD detection mechanism. Using the
14 outliers surfaced in this way, together with labeled InD data, we train a robust OOD
15 classifier. On the theoretical side, we derive error bounds that hold under mild
16 and broadly satisfiable assumptions and show that Medix attains a low error rate.
17 Extensive empirical results reinforce these guarantees: Medix consistently outper-
18 forms existing methods across the board in open-world settings, corroborating the
19 generality of our theoretical insights.

20 1 Introduction

21 Whenever machine learning models are deployed in the open world, they are routinely exposed to
22 safety and reliability challenges, and among the most pervasive is the presence of out-of-distribution
23 (OOD) data. Such OOD samples, arising from categories unseen during training, should not be
24 confidently predicted by the model. Yet neural networks are inherently vulnerable across architectures
25 and tasks, and in practice they typically lack any built-in mechanism for detecting and appropriately
26 handling OOD inputs (Nguyen et al., 2015).

27 Identifying OOD samples at inference time is at once critical and inherently difficult in virtually every
28 deployment: because models are never exposed to unknown distributions during training, they cannot
29 reliably separate OOD from in-distribution (InD) data on their own. To confront this challenge, a
30 range of recent approaches (Katz-Samuels et al., 2022a; Du et al., 2024a) have explored leveraging
31 additional “in-the-wild” data to improve OOD detection. In particular, Katz-Samuels et al. (2022a)
32 introduced a method that uses unlabeled wild data to regularize model training while still classifying
33 labeled InD data. The appeal of unlabeled wild data is its broad availability—it can be gathered
34 almost for free once a model is deployed in any operating environment. Drawing on it lets the model
35 better capture the true distribution of OOD data encountered at test time, yielding more robust OOD
36 detection across a wide variety of conditions.

Exploiting unlabeled wild data is nonetheless challenging in general, precisely because it is a complex and unknown mixture of InD and OOD data. The lack of a distinct, clean set of OOD samples complicates the design of robust OOD detection methods in any such setting, since the detector only ever observes data drawn from this mixed distribution, with no indication of whether a given sample is InD or OOD. The problem remains broadly underexplored, leaving substantial room for advances that generalize across domains. Moreover, few studies establish a formal theoretical foundation, and to the best of our knowledge Du et al. (2024a) is the only prior work to provide one for the “in-the-wild” setting.

Meanwhile, a growing line of work has demonstrated the broad effectiveness of median-based approaches in settings such as data pruning (Acharya et al., 2024). Motivated by these developments and their wide applicability, this paper sets out to answer the following question:

How can median-based methods leverage unlabeled wild data to facilitate OOD detection with theoretical guarantees?

To answer this question affirmatively, we introduce a broadly applicable median-centric perspective on OOD detection. Concretely, we propose a general median-based optimization framework and develop an algorithm, Medix, that reliably identifies OOD samples from wild data with low error rate across settings. We accompany the method with a theoretical foundation that guarantees minimal error under mild assumptions, and we further validate it through experiments that demonstrate the robustness and efficiency of the algorithm in diverse conditions. We are among the few studies to provide such a theoretical foundation for the unlabeled “in-the-wild” setting. We show that median-based filtering is robust for outlier detection in general unlabeled mixtures by bounding the fraction of InD samples incorrectly flagged as outliers. This fraction is governed by two effects: the contamination effect, which quantifies the impact of OOD points and remains manageable as long as the OOD proportion stays below 50%; and the concentration effect, which exploits the sub-Gaussian nature of InD gradients to bound deviations of InD points from their mean gradient. Together, these effects ensure that, with high probability, the number of misclassified InD points remains small, demonstrating the method’s effectiveness even in the worst case.

We benchmark our approach against two broad categories of methods: (1) those trained solely on InD data, and (2) those trained with both InD data and an auxiliary unlabeled dataset. On CIFAR-100 (Krizhevsky et al., 2009), Medix delivers a substantial improvement over the strong baseline KNN+ (Sun et al., 2022), surpassing it by an average of 40.98% in FPR95. In contrast to approaches such as Outlier Exposure (Hendrycks et al., 2019), which depend on a clean auxiliary unlabeled dataset—thereby imposing the strong distributional assumption that the auxiliary data is completely separable from the InD data—Medix attains superior results without any such assumption, making it applicable to a wider range of scenarios. Compared to WOODS (Katz-Samuels et al., 2022a), Medix reduces the average FPR95 by 1.32% on CIFAR-100 and 2.60% on CIFAR-10, with consistent gains across the individual OOD datasets.

Our key contributions are as follows:

- C1)** We propose Medix, a general median-centric greedy approach that filters outliers from unlabeled wild data and then trains an OOD detector on the identified outliers together with the InD samples. Our main contribution is the filtering stage, which is model- and domain-agnostic.
- C2)** We establish theoretical guarantees for the robustness of median-based filtering in identifying both inliers and outliers within general unlabeled mixtures. Specifically, we prove that the misclassification rates for both InD samples flagged as outliers and OOD samples retained as inliers are tightly controlled by two effects: the contamination effect, which remains bounded as long as the proportion of OOD samples is below 50%, and the concentration effect, which ensures stability in the gradient behavior of InD samples.
- C3)** We conduct an extensive evaluation of Medix across eleven InD-OOD pairs and compare its performance against 20 competitive baselines. Our results demonstrate that Medix outperforms all baselines, achieving superior performance across the board and pointing to broad applicability. We further show experimentally that Medix outlier extraction achieves a low error rate (e.g. only 12.5%; see Figure 2), corroborating our theoretical findings.

88 2 Preliminaries and Problem Setup

89 In this section, we first give a general overview of the OOD detection problem and then formally
 90 define the data setup, model architecture, loss functions, and learning goal. The formulation is
 91 intentionally kept generic so that it applies broadly across classification tasks and model families.

92 **Labeled In-Distribution Data.** Consider a generic input space $\mathcal{X}$ and label space $\mathcal{Y} = \{1, \dots, K\}$,
 93 which together define the structure of the InD data for any K -class problem. A labeled dataset
 94 $\mathcal{S}_{\text{in}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ is generated by sampling n pairs independently and identically
 95 distributed (i.i.d.) from $\mathbb{P}_{XY}$, an unknown joint probability distribution over $\mathcal{X} \times \mathcal{Y}$. The marginal
 96 distribution of $\mathbb{P}_{XY}$ on $\mathcal{X}$ is denoted as $\mathbb{P}_{\text{in}}$, representing the underlying distribution of InD inputs.
 97 We use $\mathcal{S}_{\text{in}}$ to train an InD model.

98 **Out-of-distribution detection.** We consider the general and practically ubiquitous scenario in which
 99 a model is trained on labeled InD data but is later deployed in environments that may contain OOD
 100 inputs from classes not represented in training, i.e., inputs with some label $y \notin \mathcal{Y}$. The model is
 101 expected to abstain from predicting on such OOD inputs. At inference time, the central objective is
 102 to decide whether a given input belongs to the InD distribution or arises from an OOD source—a
 103 decision that any reliably deployed model must be able to make.

104 **Unlabeled wild data.** A primary obstacle in OOD detection, common to essentially all domains, is
 105 the scarcity of labeled OOD samples. The space of possible OOD data can be enormous, rendering
 106 the collection of labeled examples both costly and impractical at scale. To address this, we bring
 107 unlabeled wild data, $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$, into our learning framework so as to faithfully mimic
 108 a broad range of real-world scenarios, following Katz-Samuels et al. (2022a). Wild data is a blend
 109 of InD and OOD samples and can be readily collected during the deployment phase of any model
 110 pre-trained on $\mathcal{S}_{\text{in}}$. Similar to Du et al. (2024a); Katz-Samuels et al. (2022a), we adopt the Huber
 111 contamination model (Huber, 1964) to characterize the marginal distribution of the wild data

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

112 where π denotes the contamination proportion and $\mathbb{P}_{\text{out}}$ captures the OOD distribution over $\mathcal{X}$. The
 113 special case $\pi = 0$ corresponds to the absence of OOD samples, which trivializes the problem; the
 114 framework is meant to cover the full range of nontrivial contamination levels $\pi \in (0, 1]$.

115 **Models and Loss Functions.** Let $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ be the InD classifier parameterized by $\phi \in \Phi$, where
 116 Φ is its parameter space. The output of f_ϕ is a soft probability distribution over the $K = |\mathcal{Y}|$ InD
 117 classes, and the formulation places no restriction on the underlying architecture. The loss function
 118 for the labeled InD data is a generic $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. For OOD detection, we introduce a separate
 119 classifier $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$, parameterized by $\theta \in \Theta$, with Θ its parameter space. The binary loss
 120 associated with g_θ is denoted $\ell_b(g_\theta(\mathbf{x}), y_b)$, where $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$; here $y_+ > 0$ represents the
 121 InD class and $y_- < 0$ the OOD class.

122 **Learning objective.** Our learning framework is designed to jointly train the OOD detector g_θ and
 123 the multi-class classifier f_ϕ , drawing on both the InD data $\mathbb{P}_{\text{in}}$ and the wild data $\mathbb{P}_{\text{wild}}$. At test time,
 124 we evaluate performance with the following standard, broadly used metrics:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{XY}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

125 where $\mathbb{I}\{\cdot\}$ denotes the indicator function, and $\mathbb{P}_{\text{out}}^{\text{test}}$ represents the OOD test data distribution.

126 3 Method: Median-Centric Framework for OOD Detection

127 In this section, we present a general learning paradigm, termed **Medix**, for OOD detection that
 128 harnesses the power of unlabeled wild data. The framework overcomes the limitations of conventional
 129 approaches that rely exclusively on InD data, and it is well-suited to the full spectrum of open-world
 130 environments in which models routinely encounter previously unseen inputs. Medix comprises two
 131 integral stages: **1) Outlier Extraction**: a filtering process that isolates candidate OOD samples from
 132 the unlabeled wild data (Section 3.1), and **2) Detector Training**: training a binary OOD detector
 133 using both InD data and the outlier candidates identified in the previous step (Section 3.2). For stage

2, we follow the protocol introduced by Du et al. (2024a). As the following sections show, this two-step methodology not only enables effective extraction of OOD data from unlabeled wild data but also provides a robust and broadly transferable foundation for deploying machine learning models in dynamic, open-world scenarios.

3.1 Extracting Candidate Outliers from the Wild Data

To isolate potential outliers from a general wild mixture $\mathcal{S}_{\text{wild}}$, our framework adopts an optimization-based approach that exploits the gradients of the model parameters. These gradients come from a classification model f_ϕ trained solely on the InD dataset $\mathcal{S}_{\text{in}}$; because the recipe depends only on such gradients, it transfers readily across models and tasks. The detailed methodology is formalized below.

Reference gradient estimation from InD data. The first step of the framework is to estimate a reference gradient from the InD dataset $\mathcal{S}_{\text{in}}$. This is done by training a classifier f_ϕ on $\mathcal{S}_{\text{in}}$ via empirical risk minimization (ERM):

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \text{ where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

where $\phi_{\mathcal{S}_{\text{in}}}$ denotes the learned parameters. Once the classifier has been trained, we compute the mean gradient $\bar{\nabla}_{\text{in}}$ as the average of the gradients of the loss function with respect to the model parameters over the InD data:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}}(\mathbf{x}_i), y_i). \quad (3)$$

Here $\bar{\nabla}_{\text{in}}$ serves as the reference gradient, allowing us to quantify how far any other data point deviates from this reference—a quantity that is meaningful regardless of the specific classifier or dataset.

Motivation. We hypothesize, as a general principle, that increasing the number of OOD samples in the wild dataset $\mathcal{S}_{\text{wild}}$ leads to greater deviation from the average InD gradient $\bar{\nabla}_{\text{in}}$. To test this hypothesis, we design an initial experiment using CIFAR-10 (Krizhevsky et al., 2009) as the InD dataset and SVHN (Netzer et al., 2011) as the OOD dataset, chosen as a representative InD–OOD pair. Specifically, $\mathcal{S}_{\text{wild}}$ consists of 10,000 samples drawn from CIFAR-10, kept disjoint from the training set used to fit the model $\phi_{\mathcal{S}_{\text{in}}}$ from which we compute $\bar{\nabla}_{\text{in}}$. We progressively add SVHN OOD samples to $\mathcal{S}_{\text{wild}}$ and track the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of the wild-data gradients:

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}_{\text{wild}}} \right) \right\|.$$

Our results, shown in Figure 1, reveal a clear and monotonic increase in the L_2 -norm deviation, supporting the hypothesis and suggesting that the underlying signal is a general one. This observation is a key motivation for the method introduced in the next section. Notably, the stopping criterion of our algorithm is derived directly from this monotonically increasing behavior: we terminate when the L_2 -norm deviation between consecutive iterations drops below a threshold ϵ , as detailed below.

Filtering potential outliers from unlabeled wild data. Motivated by the results in Figure 1, we cast the identification of the outlier subset $\mathcal{S}_{\text{out}}^*$ in $\mathcal{S}_{\text{wild}}$ as the following general optimization problem:

$$\mathcal{S}_{\text{in}}^* = \arg \min_{\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}}) \right\|, \text{ where } G_{\mathcal{S}} = \left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}}. \quad (4)$$

Here $\text{EWM}(\cdot)$ denotes the element-wise median function, and $\hat{y}_{\tilde{\mathbf{x}}_i}$ represents the predicted label for a wild sample $\tilde{\mathbf{x}}_i$. For notational simplicity, we define $G_{\mathcal{S}} = \{\nabla \ell(\tilde{\mathbf{x}}_i)\}_{i \in \mathcal{S}}$. This optimization problem seeks a subset $\mathcal{S}$ of $\mathcal{S}_{\text{wild}}$ that minimizes the distance between the EWM of the gradients and the average gradient $\bar{\nabla}_{\text{in}}$. As Figure 1 suggests, such a subset, denoted $\mathcal{S}_{\text{in}}^*$, tends to represent the InD data in $\mathcal{S}_{\text{wild}}$, in which case its complement $\mathcal{S}_{\text{out}}^* = \mathcal{S}_{\text{wild}} \setminus \mathcal{S}_{\text{in}}^*$ tends to capture the OOD data.

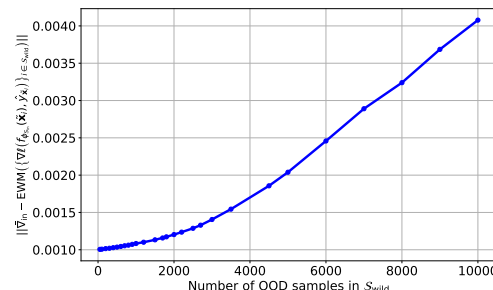


Figure 1: Distance deviation grows steadily as OOD samples are added to $\mathcal{S}_{\text{wild}}$, illustrating a general trend.

176 Solving (4) exactly can be computationally pro-
 177 hibitive, particularly as $\mathcal{S}_{\text{wild}}$ grows large. We there-
 178 fore propose a greedy approximation based on a
 179 leave-one-out strategy, outlined in Algorithm 1. The
 180 algorithm is a general iterative procedure for outlier
 181 detection from any wild dataset, driven by devi-
 182 ations in gradient information relative to the InD
 183 dataset.

184 The algorithm begins by computing the EWM of the wild data gradients at each iteration, which
 185 serves as a reference for comparing against the average gradient of the InD data $\bar{\nabla}_{\text{in}}$. We denote
 186 by d_t , the L_2 distance between the average InD gradient $\bar{\nabla}_{\text{in}}$ and the EWM gradients of the data
 187 left in the wild set, represented by $\mathcal{S}$, i.e., $d_t = \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$. The algorithm then iteratively
 188 identifies samples in $\mathcal{S}$ that incur the most significant drop in the L_2 distance with d_t when removed
 189 from $\mathcal{S}$ as OOD. Specifically, having data samples $\mathcal{S}$ left, we remove each sample $i \in \mathcal{S}$ and compute
 190 the EWM as $\text{EWM}(G_{\mathcal{S} \setminus \{i\}})$ and the distance to $\bar{\nabla}_{\text{in}}$ as $\|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$. We then find the
 191 drop in distance $\delta_i = d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$ and find the k samples with the largest drop and
 192 identify them as OOD. The algorithm repeats until there is no significant drop in δ_i or a maximum
 193 number of iterations is reached. The convergence criterion is based on the change in the L_2 distance
 194 between consecutive iterations, which must fall below a predefined threshold ϵ . This criterion is
 195 inspired by the monotonically increasing trend observed in our preliminary experiment (Figure 1): as
 196 a substantial number of OOD samples are removed, the L_2 distance gradually decreases to a small
 197 value, signaling when the algorithm should halt to avoid erroneously flagging InD samples as outliers
 198 and thereby degrading performance. This general stopping rule requires no dataset-specific tuning of
 199 the halting logic itself.

Algorithm 1 Iterative Outlier Detection via **Medix**

Require: $\bar{\nabla}_{\text{in}}, G_i = \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i})}$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$
 (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for each** $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}$, $t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\mathcal{S}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

200 3.2 Training the OOD Detector with Candidate Outliers

201 Once the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$ has been identified from the wild data, we train the OOD detector
 202 g_{θ} to maximize the separation between InD and candidate outlier data, following the protocol of Du
 203 et al. (2024a). The objective explicitly enforces separability at the decision boundary (thresholded
 204 at 0), assigning positive outputs to labeled InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ and negative outputs to candidate
 205 outliers $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$. The loss is defined as:

$$\begin{aligned} \mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_{\theta}) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^{+}(g_{\theta}) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^{-}(g_{\theta}), \\ \mathcal{L}_{\mathcal{S}_{\text{in}}}^{+}(g_{\theta}) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_{\theta}(\mathbf{x}) \leq 0\}, \\ \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^{-}(g_{\theta}) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_{\theta}(\tilde{\mathbf{x}}) > 0\}. \end{aligned} \tag{5}$$

206 To handle the non-differentiability of the 0/1 loss, we employ a differentiable sigmoid-based binary
 207 loss as a smooth surrogate, keeping the optimization tractable while remaining faithful to the original
 208 objective. The framework also incorporates the InD risk from Equation 2, safeguarding predictive
 209 accuracy on InD samples. This unified optimization markedly improves the generalization of g_{θ} ,

210 enabling robust detection of OOD samples drawn from $\mathbb{P}_{\text{out}}$ across a broad range of conditions, as
 211 our results confirm.

212 4 Theoretical Analysis

213 We now present the theoretical guarantees for Medix’s filtering stage. The following theorems provide
 214 provable upper bounds on the misclassification rates for both InD and OOD points, and they hold
 215 under mild, broadly satisfiable assumptions rather than dataset-specific ones. Together they establish
 216 the two-sided robustness of our EWM filtering. Detailed proofs appear in Appendix C.

Theorem 4.1 (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}}}_{\text{Concentration term}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

217 Theorem 4.1 shows that the inlier misclassification rate is governed by two key effects: a *concentration*
 218 *term*, which vanishes as the sample size grows and captures statistical fluctuations in the InD gradients,
 219 and a *contamination term*, which quantifies the influence of OOD points on the EWM. Crucially, the
 220 bound stays controlled for any contamination ratio $\pi < 0.5$, underscoring that median-based filtering
 221 is robust to a wide range of moderate OOD presence rather than to any single fixed level. We now
 222 turn to the complementary question: *how many true outliers escape detection and are mistakenly*
 223 *retained?* The next theorem answers this by bounding the OOD misclassification rate.

Theorem 4.2 (Outlier Misclassification Bound). *Assume that the gradients of OOD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1 - \pi}{2\pi}}_{\text{Contamination term}}.$$

224 Theorem 4.2 complements the previous result by bounding the fraction of OOD points mistakenly
 225 retained as InD. Together, Theorems 4.1 and 4.2 give a two-sided robustness guarantee for Medix
 226 filtering that applies across regimes. The inlier and outlier misclassification rates are governed by the
 227 interplay of three general effects:

228 **Contamination Effect.** For Theorem 4.1, even when the wild dataset contains a substantial fraction
 229 π of OOD points, the median-based criterion remains stable for any $\pi < 0.5$. This is captured
 230 by the $\pi/[2(1 - \pi)]$ term, which grows with π and encodes the risk that OOD gradients skew the
 231 EWM. Symmetrically, Theorem 4.2 carries a penalty $\frac{1-\pi}{2\pi}$, quantifying the difficulty of isolating
 232 OOD points when they are underrepresented—so that the guarantees span both contamination-heavy
 233 and contamination-light regimes.

234 **Concentration Effect.** Both bounds exploit the sub-Gaussian nature of the gradient coordinates, a
 235 condition met broadly in practice. For InD data, it ensures that most gradients concentrate near $\bar{\nabla}_{\text{in}}$,
 236 keeping the median stable even in finite samples. For OOD points, concentration around μ_{out} lets
 237 us quantify the risk that a misaligned OOD sample slips past the filter. These concentration effects
 238 decay as $1/\sqrt{m}$, where m is the number of samples from each class.

239 **Separation Effect.** Unique to Theorem 4.2, this effect quantifies how far the OOD mean gradient
 240 must lie from the InD mean gradient for OOD samples to be reliably rejected. The exponential term

241 $\exp(-(\Delta - \varepsilon)^2/2\sigma_{\text{out}}^2)$ captures this trade-off in general terms: the more separated the distributions,
 242 the less likely OOD gradients are to fool the filter.

Theorems 4.1 and 4.2 jointly establish that, with high probability, median filtering achieves robust separation of InD and OOD samples in general unlabeled data mixtures. The fraction of InD samples misclassified as outliers is bounded by contamination and concentration effects, while the fraction of OOD points incorrectly retained is governed by an exponential separation term, a concentration bound, and a reverse contamination effect. These results give rigorous, broadly applicable assurance that Medix minimizes both types of error under mild assumptions.

243
 244 *Remark 4.3. Sub-Gaussian Assumption for Gradient Coordinates.* The assumption that the InD-
 245 sample gradients are coordinate-wise sub-Gaussian is central to deriving concentration bounds
 246 and ensuring robust statistical estimation, and it is a mild condition that holds broadly in practice.
 247 Empirical evidence in Figure 4a supports it: the histogram of InD gradient values is bell-shaped and
 248 concentrated around the mean. This indicates light tails—a defining feature of sub-Gaussian random
 249 variables, whose tails decay exponentially so that extreme gradient values are rare—consistent with
 250 the observed histogram. Figure 4b further compares the empirical quantiles of the gradient distribution
 251 against the theoretical quantiles of a Gaussian in a Q-Q plot (Wilk and Gnanadesikan, 1968). The
 252 close alignment with the 45-degree reference line shows that the empirical gradient distribution
 253 closely resembles a Gaussian; since Gaussian variables are a subset of sub-Gaussian variables, this
 254 reinforces the assumption. *To emphasize the generality of our results, we also provide a looser*
 255 *version of these bounds that removes the sub-Gaussian assumption and holds under merely bounded*
 256 *second moments; see Theorem C.3 (Appendix C.3). While the rates degrade, the core robustness*
 257 *guarantee of Medix continues to hold in this substantially broader setting.*

258 5 Experiments

259 This section demonstrates the efficacy of Medix across a broad set of InD-OOD dataset pairs,
 260 benchmarking it against 20 widely used baselines. All experiments run on hardware equipped with
 261 NVIDIA A100-SXM4-80GB GPUs. The code needed to reproduce our results will be released upon
 262 acceptance.

263 5.1 Models, Datasets, and Baselines

264 **Datasets.** We adopt the same experimental protocol as Katz-Samuels et al. (2022a), which introduced
 265 the problem of learning OOD detectors with wild data, ensuring a fair comparison to prior work.
 266 As representative InD benchmarks we use CIFAR-10 and CIFAR-100 ($\mathbb{P}_{\text{in}}$). For OOD testing, we
 267 draw on a diverse suite of natural image datasets, including PLACES365 (Zhou et al., 2017), SVHN,
 268 TEXTURES (Cimpoi et al., 2014), and LSUN-RESIZE & LSUN-C (Yu et al., 2015) ($\mathbb{P}_{\text{out}}$), spanning
 269 a wide variety of visual domains. To simulate wild data ($\mathbb{P}_{\text{wild}}$), we mix a subset of InD data ($\mathbb{P}_{\text{in}}$)
 270 with OOD data ($\mathbb{P}_{\text{out}}$) under a default mixing parameter $\pi = 0.5$. For instance, with PLACES365
 271 as the OOD test set, we build a wild mixture by combining CIFAR with PLACES365 and test on
 272 PLACES365 as the OOD set. This procedure is repeated across every OOD dataset and baseline,
 273 giving broad coverage of InD–OOD combinations. The InD CIFAR dataset is split into two halves:
 274 the first 25,000 images train $\phi_{\mathcal{S}_{\text{in}}}$, while the remaining images form the wild mixture $\mathcal{S}_{\text{wild}}$. For
 275 gradient computation we use the penultimate-layer weights, which have been shown to be particularly
 276 informative for OOD detection in general (Huang et al., 2021a).

277 **Evaluation metrics.** We evaluate using three standard, widely reported metrics: (1) the False
 278 Positive Rate (FPR95 $\downarrow$) of OOD samples at a 95% True Positive Rate of InD samples, (2) the Area
 279 Under the Receiver Operating Characteristic Curve (AUROC $\uparrow$), and (3) InD Classification Accuracy
 280 (InD Acc $\uparrow$).

281 **Baselines.** Our comparison spans a diverse set of competitive baselines, grouped by whether they
 282 train on InD data alone or on both InD and wild data. Among methods trained exclusively on InD data
 283 ($\mathbb{P}_{\text{in}}$), we compare Medix against a broad range of established OOD detectors, including Maximum
 284 Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang et al., 2018), Mahalanobis

Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct (Sun et al., 2021), DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic et al., 2022), all of which use a model trained with softmax cross-entropy loss. For broader coverage we also include contrastive-loss methods such as CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022). Among methods that leverage both InD and wild data, we compare against Outlier Exposure (OE) (Hendrycks et al., 2019) and energy-regularization learning (Liu et al., 2020b), which regularize training by encouraging lower confidence or higher energy on outlier data. We further compare with WOODS (Katz-Samuels et al., 2022a), which introduced the notion of wild unlabeled data and uses it for OOD detection via a constrained optimization approach. Finally, to make the evaluation as thorough and representative as possible, we add more recent baselines, including CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024).

5.2 Experimental Setup

Following WOODS (Katz-Samuels et al., 2022a), we use a Wide ResNet architecture (Zagoruyko, 2016) with 40 layers and a width factor of 2 for the InD classifier $\phi_{S_{in}}$, though the framework is agnostic to this choice. It is trained with stochastic gradient descent using momentum 0.9, weight decay 0.0005, and an initial learning rate of 0.1, for 100 epochs with cosine learning-rate decay, a batch size of 128, and a dropout rate of 0.3. The Medix hyperparameters ϵ and k are selected from the sets $\{5e-5, 5e-4, 5e-3, 5e-2\}$ and $\{4k, 7k, 10k, 20k\}$, respectively, chosen to accommodate a range of dataset sizes while maximizing OOD performance. For the OOD classifier g_θ , we initialize it with the pre-trained InD classifier $\phi_{S_{in}}$ and append a linear layer that performs binary classification on the penultimate-layer features. Its learning rate is set to 0.001, and fine-tuning runs for 100 epochs following Equation (5). We combine the binary classification loss with the InD classification loss, weighting the binary component by 10. All remaining training parameters match those used for $\phi_{S_{in}}$.

Table 1: OOD detection performance of Medix and baselines with CIFAR-10 as InD data, reported across a diverse suite of OOD datasets. Results are averaged over five runs; the best results in each column are highlighted in **bold**.

Methods	OOD Datasets												InD Acc $\uparrow$	
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average			
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$		
	Using F_{in} only													
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84	
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84	
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84	
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84	
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84	
ReAct	40.76	89.57	41.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84	
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84	
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84	
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17	
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19	
	Using F_{in} and F_{wild}													
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12	
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24	
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74	
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58	
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64	

Table 2: OOD detection performance of Medix and baselines with CIFAR-100 as InD data, reported across a diverse suite of OOD datasets. Results are averaged over five runs; the best results in each column are highlighted in **bold**.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
	Using F_{in} only												
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55	75.96
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38	75.96
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03	75.96
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69	75.96
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08	75.96
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01	75.96
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39	75.96
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77	75.96
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44	69.90
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29	73.78
Using F_{in} and F_{wild}													
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17	73.38
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51	72.76
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73	73.91
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96	73.33
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10	± 0.83

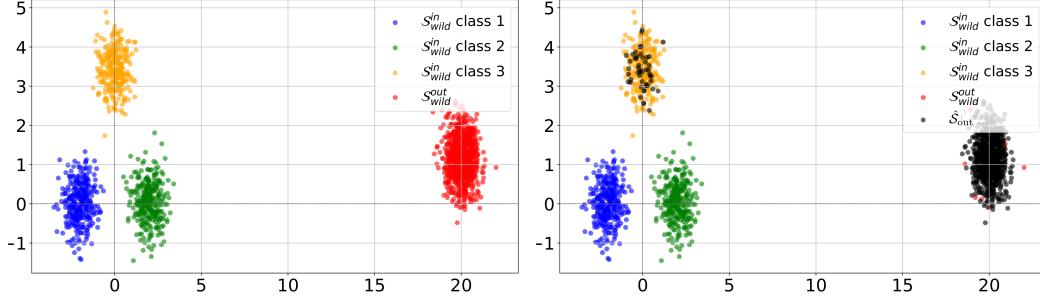


Figure 2: Illustrative example of Medix applied to unlabeled wild data. (a) Setup of the InD data $\mathcal{S}_{\text{wild}}^{\text{in}}$ and OOD data $\mathcal{S}_{\text{wild}}^{\text{out}}$ in the wild, with inliers sampled from three multivariate Gaussian distributions. (b) Outliers $\hat{\mathcal{S}}_{\text{out}}$ filtered by Medix (in black); the error rate at which $\hat{\mathcal{S}}_{\text{out}}$ contains InD data $\mathcal{S}_{\text{wild}}^{\text{in}}$ is only 12.5%.

5.3 Results

Our main results on CIFAR-100 are presented in Table 2, where Medix consistently outperforms all OOD detection baselines. Two broad observations stand out. (1) Methods trained on both InD and wild data substantially outperform those trained on InD data alone: relative to KNN+, Medix reduces FPR95 by 52.31% on PLACES365 and 38.24% on TEXTURES, underscoring the general value of incorporating in-the-wild data for regularization. (2) Medix further surpasses competitive methods that also use wild data ($\mathbb{P}_{\text{wild}}$): on CIFAR-100 it attains an average FPR95 of 5.42%, a 1.32% improvement over WOODS, while maintaining a competitive InD accuracy of 73.33%. The small accuracy gap arises because our method is trained on 25,000 labeled InD samples, whereas baselines that do not leverage wild data use the full CIFAR-100 training set of 50,000 samples. Results for CIFAR-10 appear in Table 1, where Medix again surpasses every baseline. It outperforms WOODS by 7.21% on PLACES365 and 5.25% on TEXTURES in FPR95, illustrating that its effectiveness at detecting OOD samples carries over across InD settings.

A representative visual example of Medix. We further examine the behavior of Algorithm 1 (Outlier Extraction) when extracting OOD samples from wild data $\mathcal{S}_{\text{wild}}$. To make the mechanism transparent, we design a deliberately simple experiment on 2-dimensional synthetic data, which serves as a controlled illustration of the general principle at work. We generate the InD data by sampling from three multivariate Gaussian distributions corresponding to three classes, with mean vectors $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$, respectively, and a shared covariance matrix $0.25 \cdot I$. For the OOD data we use a multivariate Gaussian $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. As Figure 2 shows, our method successfully identifies outliers, the majority of which align with the ground truth: Medix flags 87.5% of the true OOD samples as outliers, underscoring the robustness of its outlier extraction.

Additional studies and insights. Due to space constraints, we defer further experiments and insights to Appendix A. These span a wide range of conditions and include (1) ablation studies on hyperparameter selection and sensitivity analysis (Appendix A.2), showing Medix’s strong robustness across hyperparameter choices; (2) a comparison between EWM and geometric median (Appendix A.1), showing that EWM is more sensitive to distributional shifts and therefore a more effective and reliable filtering choice in our method; (3) a comprehensive comparison against methods with competitive contrastive learning objectives (Appendix A.3), where Medix outperforms even these strong baselines by a significant margin; (4) an evaluation on large-scale data under the more demanding unseen-OOD setting, where $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$ (Appendix A.4), where Medix again outperforms baselines by a wide margin; (5) an assessment of the computation and memory efficiency of Medix (Appendix A.6); (6) an analysis of the impact of pseudo-label quality (Appendix A.5), showing that our method is resilient to noisy or low-confidence labels; and (7) a comparison with semi-supervised open-set recognition methods (Appendix A.7).

6 Related Work

Interest in OOD detection has grown rapidly and now spans a wide array of tasks and settings (Fort et al., 2021; Yang et al., 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al., 2023; Djuricic et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al., 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024; Abbas et al., 2025). A broad family of approaches detects OOD data through scoring functions that assess the data distribution, including distance-based methods (Lee et al., 2018b; Tack et al., 2020b; Ren et al., 2021; Sehwag et al., 2021; Sun et al., 2022; Du et al., 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based scores (Huang et al., 2021b), energy-based scores (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023), confidence-based approaches (Bendale and Boult, 2016; Hendrycks and Gimpel, 2017; Liang et al., 2018), and Bayesian methods (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin and Gales, 2019; Wen et al., 2020; Kristiadi et al., 2020).

A complementary line of work performs OOD detection through regularization during training (Malinin and Gales, 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein, 2020; Jeong and Kim, 2020; Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b, 2023; Wang et al., 2023a). For example, models can be regularized to reduce confidence (Lee et al., 2017; Hendrycks et al., 2019) or increase energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on OOD data. Most such regularization methods, however, assume access to an auxiliary OOD dataset—an assumption that limits their applicability in the wild.

Several studies (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023) relax the reliance on labeled data alone by incorporating unlabeled wild data (Katz-Samuels et al., 2022a; Geng et al., 2025), though without proposing a clear outlier-detection mechanism. In contrast, Du et al. (2024a,b) introduced an explicit outlier filtering method, but their thresholding technique differs fundamentally from ours, which uses a new median-centric approach to detect outliers. Moreover, Katz-Samuels et al. (2022a); Du et al. (2024a) operate under a batch-level mixing assumption, where each batch contains a fixed ratio of InD and OOD samples. In many practical, large-scale outsourced datasets such structured mixing is unavailable—data is mixed randomly throughout the dataset—and our method addresses this more general regime by supporting dataset-level mixing without relying on batch-level structure. A further body of work leverages positive-unlabeled learning, training classifiers from positive and/or unlabeled data (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu et al., 2016; Gong et al., 2018; Chapel et al., 2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al., 2022; Zhao et al., 2022; Acharya et al., 2022). A key distinction is that these methods focus solely on distinguishing $\mathbb{P}_{\text{out}}$ from $\mathbb{P}_{\text{in}}$ and do not simultaneously train an OOD classifier. By comparison, we propose a general median-centric method for identifying outliers in unlabeled data with provably low error rates.

7 Conclusions

In this work, we introduced Medix, a general median-centric framework for OOD detection that leverages unlabeled in-the-wild data. Exploiting the inherent robustness of the median operation, Medix effectively filters outliers from mixed unlabeled data of essentially any composition, enabling the training of a reliable OOD detector. Our theoretical analysis established provable bounds on the inlier misclassification rate, showing that Medix remains robust even under significant OOD contamination (up to 50%), with errors controlled by broadly applicable sub-Gaussian concentration and contamination effects. We further provided complementary theoretical limits on the OOD misclassification rate, characterizing when OOD samples can be reliably isolated under clear separation conditions. Empirical validation across a diverse range of benchmarks confirmed Medix’s superiority over 20 baselines: it reduced the average FPR95 by 40.98% relative to strong baselines such as KNN+, outperformed state-of-the-art methods including WOODS and DRL, and achieved an outlier extraction error rate as low as 12.5%. Taken together, these theoretical and empirical results point to Medix as a broadly applicable foundation for OOD detection in open-world deployments.

References

Abbas, M., Azmat, M., Horesh, R., and Yurochkin, M. (2025). Out-of-distribution detection using synthetic data generation. In *Second Conference on Language Modeling*.

396 Acharya, A., Dhillon, I. S., and Sanghavi, S. (2024). Geometric median (gm) matching for robust data pruning.
397 *arXiv preprint arXiv:2406.17188*.

398 Acharya, A., Sanghavi, S., Jing, L., Bhushanam, B., Choudhary, D., Rabbat, M., and Dhillon, I. (2022). Positive
399 unlabeled contrastive learning. *arXiv preprint arXiv:2206.01206*.

400 Bai, H., Canal, G., Du, X., Kwon, J., Nowak, R. D., and Li, Y. (2023). Feed two birds with one scone: Exploiting
401 wild data for both out-of-distribution generalization and detection. In *International Conference on Machine*
402 *Learning*, pages 1454–1471. PMLR.

403 Bendale, A. and Boulton, T. E. (2016). Towards open set deep networks. In *Proceedings of the IEEE conference*
404 *on computer vision and pattern recognition*, pages 1563–1572.

405 Chapel, L., Alaya, M. Z., and Gasso, G. (2020). Partial optimal transport with applications on positive-unlabeled
406 learning. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural*
407 *Information Processing Systems*, volume 33, pages 2903–2913. Curran Associates, Inc.

408 Cimpoi, M., Maji, S., Kokkinos, I., Mohamed, S., and Vedaldi, A. (2014). Describing textures in the wild. In
409 *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3606–3613.

410 Djurisic, A., Bozanic, N., Ashok, A., and Liu, R. (2022). Extremely simple activation shaping for out-of-
411 distribution detection. *arXiv preprint arXiv:2209.09858*.

412 Djurisic, A., Bozanic, N., Ashok, A., and Liu, R. (2023). Extremely simple activation shaping for out-of-
413 distribution detection. In *The Eleventh International Conference on Learning Representations*.

414 Du, X., Fang, Z., Diakonikolas, I., and Li, Y. (2024a). How does unlabeled data provably help out-of-distribution
415 detection? In *The Twelfth International Conference on Learning Representations*.

416 Du, X., Ghosh, R., Sim, R., Salem, A., Carvalho, V., Lawton, E., Li, Y., and Stokes, J. W. (2024b). Vlmguard:
417 Defending vlms against malicious prompts via unlabeled data. *arXiv preprint arXiv:2410.00296*.

418 Du, X., Gozum, G., Ming, Y., and Li, Y. (2022a). Siren: Shaping representations for detecting out-of-distribution
419 objects. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in*
420 *Neural Information Processing Systems*, volume 35, pages 20434–20449. Curran Associates, Inc.

421 Du, X., Sun, Y., Zhu, J., and Li, Y. (2023). Dream the impossible: Outlier imagination with diffusion models.
422 In Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine, S., editors, *Advances in Neural*
423 *Information Processing Systems*, volume 36, pages 60878–60901. Curran Associates, Inc.

424 Du, X., Wang, X., Gozum, G., and Li, Y. (2022b). Unknown-aware object detection: Learning what you don’t
425 know from videos in the wild. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern*
426 *Recognition*, pages 13678–13688.

427 Du, X., Wang, Z., Cai, M., and Li, S. (2022c). Towards unknown-aware learning with virtual outlier synthesis.
428 In *International Conference on Learning Representations*.

429 Fan, Y., Kukleva, A., Dai, D., and Schiele, B. (2023). Ssb: Simple but strong baseline for boosting performance
430 of open-set semi-supervised learning. In *Proceedings of the IEEE/CVF International Conference on Computer*
431 *Vision (ICCV)*, pages 16068–16078.

432 Fang, Z., Li, Y., Lu, J., Dong, J., Han, B., and Liu, F. (2022). Is out-of-distribution detection learnable? In
433 Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in Neural*
434 *Information Processing Systems*, volume 35, pages 37199–37213. Curran Associates, Inc.

435 Fort, S., Ren, J., and Lakshminarayanan, B. (2021). Exploring the limits of out-of-distribution detection.
436 *Advances in Neural Information Processing Systems*, 34:7068–7081.

437 Gal, Y. and Ghahramani, Z. (2016). Dropout as a bayesian approximation: Representing model uncertainty
438 in deep learning. In Balcan, M. F. and Weinberger, K. Q., editors, *Proceedings of The 33rd International*
439 *Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 1050–
440 1059, New York, New York, USA. PMLR.

441 Galil, I., Dabbah, M., and El-Yaniv, R. (2023). A framework for benchmarking class-out-of-distribution detection
442 and its application to imagenet. In *The Eleventh International Conference on Learning Representations*.

443 Garg, S., Balakrishnan, S., and Lipton, Z. (2022). Domain adaptation under open set label shift. In Koyejo,
444 S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in Neural Information*
445 *Processing Systems*, volume 35, pages 22531–22546. Curran Associates, Inc.

446 Garg, S., Wu, Y., Smola, A. J., Balakrishnan, S., and Lipton, Z. (2021). Mixture proportion estimation and
447 pu learning: a modern approach. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan,
448 J. W., editors, *Advances in Neural Information Processing Systems*, volume 34, pages 8532–8544. Curran
449 Associates, Inc.

450 Geifman, Y. and El-Yaniv, R. (2019). SelectiveNet: A deep neural network with an integrated reject option. In
451 Chaudhuri, K. and Salakhutdinov, R., editors, *Proceedings of the 36th International Conference on Machine
452 Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2151–2159. PMLR.

453 Geng, C., Li, Q., Wang, X., Liang, D., Chen, S., and Yuen, P. C. (2025). Lod: Loss-difference ood detection by
454 intentionally label-noisifying unlabeled wild data. *arXiv preprint arXiv:2505.12952*.

455 Ghosal, S. S., Sun, Y., and Li, Y. (2024). How to overcome curse-of-dimensionality for out-of-distribution
456 detection? In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 19849–19857.

457 Gong, T., Wang, G., Ye, J., Xu, Z., and Lin, M. (2018). Margin based pu learning. In *Proceedings of the AAAI
458 Conference on Artificial Intelligence*, volume 32.

459 Hang, J.-Y. and Zhang, M.-L. (2024). Binary decomposition: A problem transformation perspective for open-set
460 semi-supervised learning. In Salakhutdinov, R., Koltner, Z., Heller, K., Weller, A., Oliver, N., Scarlett, J., and
461 Berkenkamp, F., editors, *Proceedings of the 41st International Conference on Machine Learning*, volume 235
462 of *Proceedings of Machine Learning Research*, pages 17505–17518. PMLR.

463 He, R., Li, R., Han, Z., Yang, X., and Yin, Y. (2023). Topological structure learning for weakly-supervised
464 out-of-distribution detection. In *Proceedings of the 31st ACM International Conference on Multimedia*, pages
465 4858–4866.

466 Hein, M., Andriushchenko, M., and Bitterwolf, J. (2019). Why relu networks yield high-confidence predictions
467 far away from the training data and how to mitigate the problem. In *Proceedings of the IEEE/CVF conference
468 on computer vision and pattern recognition*, pages 41–50.

469 Hendrycks, D. and Gimpel, K. (2016). A baseline for detecting misclassified and out-of-distribution examples in
470 neural networks. *arXiv preprint arXiv:1610.02136*.

471 Hendrycks, D. and Gimpel, K. (2017). A baseline for detecting misclassified and out-of-distribution examples in
472 neural networks. In *International Conference on Learning Representations*.

473 Hendrycks, D., Mazeika, M., and Dietterich, T. (2019). Deep anomaly detection with outlier exposure. In
474 *International Conference on Learning Representations*.

475 Hsieh, C.-J., Natarajan, N., and Dhillon, I. (2015). Pu learning for matrix completion. In Bach, F. and Blei, D.,
476 editors, *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings
477 of Machine Learning Research*, pages 2445–2453, Lille, France. PMLR.

478 Huang, R., Geng, A., and Li, Y. (2021a). On the importance of gradients for detecting distributional shifts in the
479 wild. *Advances in Neural Information Processing Systems*, 34:677–689.

480 Huang, R., Geng, A., and Li, Y. (2021b). On the importance of gradients for detecting distributional shifts in the
481 wild. In Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W., editors, *Advances in Neural Information
482 Processing Systems*.

483 Huber, P. J. (1964). Robust estimation of a location parameter. *The Annals of Mathematical Statistics*, 35(1):73–
484 101.

485 Jeong, T. and Kim, H. (2020). Ood-maml: Meta-learning for few-shot out-of-distribution detection and
486 classification. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in
487 Neural Information Processing Systems*, volume 33, pages 3907–3916. Curran Associates, Inc.

488 Katz-Samuels, J., Nakhleh, J. B., Nowak, R., and Li, Y. (2022a). Training ood detectors in their natural habitats.
489 In *International Conference on Machine Learning*, pages 10848–10865. PMLR.

490 Katz-Samuels, J., Nakhleh, J. B., Nowak, R., and Li, Y. (2022b). Training OOD detectors in their natural
491 habitats. In Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S., editors, *Proceedings
492 of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning
493 Research*, pages 10848–10865. PMLR.

494 Kim, S., Kim, D., Cho, M., and Kwak, S. (2020). Proxy anchor loss for deep metric learning. In *2020 IEEE/CVF
495 Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3235–3244.

496 Kristiadi, A., Hein, M., and Hennig, P. (2020). Being bayesian, even just a bit, fixes overconfidence in relu
497 networks. In *International conference on machine learning*, pages 5436–5446. PMLR.

498 Krizhevsky, A., Hinton, G., et al. (2009). Learning multiple layers of features from tiny images.

499 Lakshminarayanan, B., Pritzel, A., and Blundell, C. (2017). Simple and scalable predictive uncertainty estimation
500 using deep ensembles. In Guyon, I., Luxburg, U. V., Bengio, S., Wallach, H., Fergus, R., Vishwanathan, S.,
501 and Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates,
502 Inc.

503 Lee, K., Lee, H., Lee, K., and Shin, J. (2017). Training confidence-calibrated classifiers for detecting out-of-
504 distribution samples. *arXiv preprint arXiv:1711.09325*.

505 Lee, K., Lee, K., Lee, H., and Shin, J. (2018a). A simple unified framework for detecting out-of-distribution
506 samples and adversarial attacks. *Advances in neural information processing systems*, 31.

507 Lee, K., Lee, K., Lee, H., and Shin, J. (2018b). A simple unified framework for detecting out-of-distribution
508 samples and adversarial attacks. In Bengio, S., Wallach, H., Larochelle, H., Grauman, K., Cesa-Bianchi, N.,
509 and Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 31. Curran Associates,
510 Inc.

511 Letouzey, F., Denis, F., and Gilleron, R. (2000). Learning from positive and unlabeled examples. In *International
512 Conference on Algorithmic Learning Theory*, pages 71–85. Springer.

513 Liang, S., Li, Y., and Srikant, R. (2018). Enhancing the reliability of out-of-distribution image detection in
514 neural networks. In *International Conference on Learning Representations*.

515 Liu, J., Lin, Z., Padhy, S., Tran, D., Bedrax Weiss, T., and Lakshminarayanan, B. (2020a). Simple and principled
516 uncertainty estimation with deterministic deep learning via distance awareness. In Larochelle, H., Ranzato,
517 M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing Systems*,
518 volume 33, pages 7498–7512. Curran Associates, Inc.

519 Liu, W., Wang, X., Owens, J., and Li, Y. (2020b). Energy-based out-of-distribution detection. *Advances in
520 neural information processing systems*, 33:21464–21475.

521 Liu, W., Wang, X., Owens, J., and Li, Y. (2020c). Energy-based out-of-distribution detection. In Larochelle,
522 H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing
523 Systems*, volume 33, pages 21464–21475. Curran Associates, Inc.

524 Maddox, W. J., Izmailov, P., Garipov, T., Vetrov, D. P., and Wilson, A. G. (2019). A simple baseline for bayesian
525 uncertainty in deep learning. In Wallach, H., Larochelle, H., Beygelzimer, A., d’Alché-Buc, F., Fox, E., and
526 Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.

527 Malinin, A. and Gales, M. (2018). Predictive uncertainty estimation via prior networks. In Bengio, S., Wallach,
528 H., Larochelle, H., Grauman, K., Cesa-Bianchi, N., and Garnett, R., editors, *Advances in Neural Information
529 Processing Systems*, volume 31. Curran Associates, Inc.

530 Malinin, A. and Gales, M. (2019). Reverse kl-divergence training of prior networks: Improved uncertainty and
531 adversarial robustness. In Wallach, H., Larochelle, H., Beygelzimer, A., d’Alché-Buc, F., Fox, E., and Garnett,
532 R., editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.

533 Meinke, A. and Hein, M. (2020). Towards neural networks that provably know when they don’t know. In
534 *International Conference on Learning Representations*.

535 Ming, Y., Fan, Y., and Li, Y. (2022). POEM: Out-of-distribution detection with posterior sampling. In
536 Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S., editors, *Proceedings of the 39th
537 International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*,
538 pages 15650–15665. PMLR.

539 Ming, Y. and Li, Y. (2024). How does fine-tuning impact out-of-distribution detection for vision-language
540 models? *International Journal of Computer Vision*, 132(2):596–609.

541 Ming, Y., Sun, Y., Dia, O., and Li, Y. (2023). How to exploit hyperspherical embeddings for out-of-distribution
542 detection? In *The Eleventh International Conference on Learning Representations*.

543 Netzer, Y., Wang, T., Coates, A., Bissacco, A., Wu, B., Ng, A. Y., et al. (2011). Reading digits in natural images
544 with unsupervised feature learning. In *NIPS workshop on deep learning and unsupervised feature learning*,
545 volume 2011, page 4. Granada.

546 Nguyen, A., Yosinski, J., and Clune, J. (2015). Deep neural networks are easily fooled: High confidence
547 predictions for unrecognizable images. In *Proceedings of the IEEE conference on computer vision and pattern
548 recognition*, pages 427–436.

549 Niu, G., Du Plessis, M. C., Sakai, T., Ma, Y., and Sugiyama, M. (2016). Theoretical comparisons of positive-
550 unlabeled learning against positive-negative learning. *Advances in neural information processing systems*,
551 29.

552 Peng, B., Luo, Y., Zhang, Y., Li, Y., and Fang, Z. (2024). Conjnorm: Tractable density estimation for out-of-
553 distribution detection. In *The Twelfth International Conference on Learning Representations*.

554 Plessis, M. D., Niu, G., and Sugiyama, M. (2015). Convex formulation for learning from positive and unlabeled
555 data. In Bach, F. and Blei, D., editors, *Proceedings of the 32nd International Conference on Machine Learning*,
556 volume 37 of *Proceedings of Machine Learning Research*, pages 1386–1394, Lille, France. PMLR.

557 Ren, J., Fort, S., Liu, J., Roy, A. G., Padhy, S., and Lakshminarayanan, B. (2021). A simple fix to mahalanobis
558 distance for improving near-ood detection. *arXiv preprint arXiv:2106.09022*.

559 Ren, J., Luo, J., Zhao, Y., Krishna, K., Saleh, M., Lakshminarayanan, B., and Liu, P. J. (2022). Out-of-distribution
560 detection and selective generation for conditional language models. In *The Eleventh International Conference
561 on Learning Representations*.

562 Saito, K., Kim, D., and Saenko, K. (2021). Openmatch: Open-set semi-supervised learning with open-set
563 consistency regularization. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W.,
564 editors, *Advances in Neural Information Processing Systems*, volume 34, pages 25956–25967. Curran
565 Associates, Inc.

566 Sehwag, V., Chiang, M., and Mittal, P. (2021). {SSD}: A unified framework for self-supervised outlier detection.
567 In *International Conference on Learning Representations*.

568 Sun, Y., Guo, C., and Li, Y. (2021). React: Out-of-distribution detection with rectified activations. *Advances in
569 Neural Information Processing Systems*, 34:144–157.

570 Sun, Y. and Li, Y. (2022). Dice: Leveraging sparsification for out-of-distribution detection. In *European
571 Conference on Computer Vision*, pages 691–708. Springer.

572 Sun, Y., Ming, Y., Zhu, X., and Li, Y. (2022). Out-of-distribution detection with deep nearest neighbors. In
573 *International Conference on Machine Learning*, pages 20827–20840. PMLR.

574 Tack, J., Mo, S., Jeong, J., and Shin, J. (2020a). Csi: Novelty detection via contrastive learning on distributionally
575 shifted instances. *Advances in neural information processing systems*, 33:11839–11852.

576 Tack, J., Mo, S., Jeong, J., and Shin, J. (2020b). Csi: Novelty detection via contrastive learning on distributionally
577 shifted instances. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in
578 Neural Information Processing Systems*, volume 33, pages 11839–11852. Curran Associates, Inc.

579 Tao, L., Du, X., Zhu, J., and Li, Y. (2023). Non-parametric outlier synthesis. In *The Eleventh International
580 Conference on Learning Representations*.

581 Uppaal, R., Hu, J., and Li, Y. (2023). Is fine-tuning needed? pre-trained language models are near perfect
582 for out-of-domain detection. In Rogers, A., Boyd-Graber, J., and Okazaki, N., editors, *Proceedings of the
583 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages
584 12813–12832, Toronto, Canada. Association for Computational Linguistics.

585 Van Amersfoort, J., Smith, L., Teh, Y. W., and Gal, Y. (2020). Uncertainty estimation using a single deep
586 deterministic neural network. In *International conference on machine learning*, pages 9690–9700. PMLR.

587 Wang, H., Li, Z., Feng, L., and Zhang, W. (2022a). Vim: Out-of-distribution with virtual-logit matching. In
588 *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4921–4930.

589 Wang, H., Liu, W., Bocchieri, A., and Li, Y. (2021). Can multi-label classification networks know what they
590 don’t know? *Advances in Neural Information Processing Systems*, 34:29074–29087.

591 Wang, Q., Fang, Z., Zhang, Y., Liu, F., Li, Y., and Han, B. (2023a). Learning to augment distributions for
592 out-of-distribution detection. In Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine,
593 S., editors, *Advances in Neural Information Processing Systems*, volume 36, pages 73274–73286. Curran
594 Associates, Inc.

595 Wang, Q., Liu, F., Zhang, Y., Zhang, J., Gong, C., Liu, T., and Han, B. (2022b). Watermarking for out-of-
596 distribution detection. *Advances in Neural Information Processing Systems*, 35:15545–15557.

597 Wang, Q., Ye, J., Liu, F., Dai, Q., Kalander, M., Liu, T., HAO, J., and Han, B. (2023b). Out-of-distribution
598 detection with implicit outlier transformation. In *The Eleventh International Conference on Learning*
599 *Representations*.

600 Wang, Y., Zou, J., Lin, J., Ling, Q., Pan, Y., Yao, T., and Mei, T. (2022c). Out-of-distribution detection via
601 conditional kernel independence model. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K.,
602 and Oh, A., editors, *Advances in Neural Information Processing Systems*, volume 35, pages 36411–36425.
603 Curran Associates, Inc.

604 Wang, Z., Xiang, L., Huang, L., Mao, J., Xiao, L., and Yamasaki, T. (2024). Scomatch: Alleviating overtrusting
605 in open-set semi-supervised learning. In *European Conference on Computer Vision*, pages 217–233. Springer.

606 Wei, H., Xie, R., Cheng, H., Feng, L., An, B., and Li, Y. (2022). Mitigating neural network overconfidence
607 with logit normalization. In Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S.,
608 editors, *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings*
609 *of Machine Learning Research*, pages 23631–23644. PMLR.

610 Wen, Y., Tran, D., and Ba, J. (2020). Batchensemble: an alternative approach to efficient ensemble and lifelong
611 learning. In *International Conference on Learning Representations*.

612 Wilk, M. B. and Gnanadesikan, R. (1968). Probability plotting methods for the analysis of data. *Biometrika*,
613 55(1):1–17.

614 Wu, Q., Chen, Y., Yang, C., and Yan, J. (2023). Energy-based out-of-distribution detection for graph neural
615 networks. In *The Eleventh International Conference on Learning Representations*.

616 Xu, D. and Denil, M. (2021). Positive-unlabeled reward learning. In *Conference on Robot Learning*, pages
617 205–219. PMLR.

618 Yang, J., Wang, H., Feng, L., Yan, X., Zheng, H., Zhang, W., and Liu, Z. (2021). Semantically coherent
619 out-of-distribution detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*,
620 pages 8301–8309.

621 Yang, J., Wang, P., Zou, D., Zhou, Z., Ding, K., Peng, W., Wang, H., Chen, G., Li, B., Sun, Y., et al.
622 (2022). Openood: Benchmarking generalized out-of-distribution detection. *Advances in Neural Information*
623 *Processing Systems*, 35:32598–32611.

624 Yang, J., Zhou, K., Li, Y., and Liu, Z. (2024). Generalized out-of-distribution detection: A survey. *International*
625 *Journal of Computer Vision*, 132(12):5635–5662.

626 Yu, F., Seff, A., Zhang, Y., Song, S., Funkhouser, T., and Xiao, J. (2015). Lsun: Construction of a large-scale
627 image dataset using deep learning with humans in the loop. *arXiv preprint arXiv:1506.03365*.

628 Zagoruyko, S. (2016). Wide residual networks. *arXiv preprint arXiv:1605.07146*.

629 Zhang, J., Yang, J., Wang, P., Wang, H., Lin, Y., Zhang, H., Sun, Y., Du, X., Zhou, K., Zhang, W., et al. (2023).
630 Openood v1. 5: Enhanced benchmark for out-of-distribution detection. *arXiv preprint arXiv:2306.09301*.

631 Zhang, Y., Lu, J., Peng, B., Fang, Z., and ming Cheung, Y. (2024). Learning to shape in-distribution feature space
632 for out-of-distribution detection. In *The Thirty-eighth Annual Conference on Neural Information Processing*
633 *Systems*.

634 Zhao, Y., Xu, Q., Jiang, Y., Wen, P., and Huang, Q. (2022). Dist-pu: Positive-unlabeled learning from a label
635 distribution perspective. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern*
636 *Recognition*, pages 14461–14470.

637 Zheng, H., Wang, Q., Fang, Z., Xia, X., Liu, F., Liu, T., and Han, B. (2023). Out-of-distribution detection
638 learning with unreliable out-of-distribution sources. *Advances in Neural Information Processing Systems*,
639 36:72110–72123.

640 Zhou, B., Lapedriza, A., Khosla, A., Oliva, A., and Torralba, A. (2017). Places: A 10 million image database for
641 scene recognition. *IEEE transactions on pattern analysis and machine intelligence*, 40(6):1452–1464.

642 Zhou, Z., Guo, L.-Z., Cheng, Z., Li, Y.-F., and Pu, S. (2021). Step: Out-of-distribution detection in the presence
643 of limited in-distribution labeled data. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan,
644 J. W., editors, *Advances in Neural Information Processing Systems*, volume 34, pages 29168–29180. Curran
645 Associates, Inc.

- 646 Zhu, J., Geng, Y., Yao, J., Liu, T., Niu, G., Sugiyama, M., and Han, B. (2023). Diversified outlier exposure
647 for out-of-distribution detection via informative extrapolation. *Advances in Neural Information Processing*
648 *Systems*, 36:22702–22734.
- 649 Zhu, Y., Chen, Y., Xie, C., Li, X., Zhang, R., Xue', H., Tian, X., zheng, b., and Chen, Y. (2022). Boosting
650 out-of-distribution detection with typical features. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave,
651 D., Cho, K., and Oh, A., editors, *Advances in Neural Information Processing Systems*, volume 35, pages
652 20758–20769. Curran Associates, Inc.

A Additional Studies

A.1 Element-wise Median vs Geometric Median

A natural question is why we chose the element-wise median over the geometric median (Acharya et al., 2024) for outlier detection in Medix, given that both are broadly used robust estimators. To answer this, we conducted a preliminary experiment using CIFAR-10 as the InD dataset and SVHN as the OOD dataset, taken as a representative pair. Specifically, $\mathcal{S}_{\text{wild}}$ consists of 5k samples drawn from CIFAR-10, kept disjoint from the training set used to fit the model $\phi_{\mathcal{S}_{\text{in}}}$ from which we compute $\bar{\nabla}_{\text{in}}$. We incrementally add SVHN OOD samples to $\mathcal{S}_{\text{wild}}$ and track both the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median of the wild-data gradients and the proportion of OOD samples removed. The results, shown in Figure 3, indicate that for the same number of OOD samples in the wild set the element-wise median flags a substantially higher proportion of OOD samples as outliers than the geometric median. This suggests that the element-wise median is more sensitive to distributional shifts, making it a more effective and reliable filtering choice across settings in our method.

Table 3: Effect of the hyperparameters ϵ and k on OOD detection performance.

Method	FPR95↓	ϵ	k
DICE	88.35	—	—
ASH	21.36	—	—
CSI	64.70	—	—
KNN+	32.21	—	—
OE	2.86	—	—
Energy	2.71	—	—
Medix	0.16	0.005	20000
Medix	0.20	0.0005	20000
Medix	0.68	0.005	10000

A.2 Hyperparameter Selection and Sensitivity Analysis

We conducted an ablation study to assess the sensitivity of Medix to the hyperparameters ϵ and k , using CIFAR-100 as the InD dataset and SVHN as the OOD dataset. As shown in Table 3, Medix is strongly robust to variations in ϵ and k . In particular, it attains a remarkably low FPR95 of 0.16 with $\epsilon = 0.005$ and $k = 20000$. Even under substantial changes—reducing ϵ to 0.0005 or halving k to 10000—performance stays competitive (FPR95 of 0.20 and 0.68, respectively), reflecting graceful degradation rather than a sharp drop. These results indicate that, while optimal tuning helps, Medix retains its effectiveness across a wide range of hyperparameter choices and continues to surpass the baselines. This insensitivity to exact tuning makes Medix a dependable choice for real-world deployments across domains, where exhaustive tuning is often infeasible.

A.3 Comprehensive comparison with recent competitive methods

To broaden the scope of our evaluation, we extend the comparison to competitive baselines built on diverse contrastive learning objectives, such as SSD+ (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020), and CIDER Ming et al. (2023). Although these methods rely on contrastive objectives, which we do not, we include them for a comprehensive comparison. We also add more recent methods, including CONJ (Peng et al., 2024), DRL (Zhang et al., 2024), Vim (Wang et al., 2022a), and VOS (Du et al., 2022c), for a still more thorough evaluation. For this experiment we use CIFAR-100 as the InD

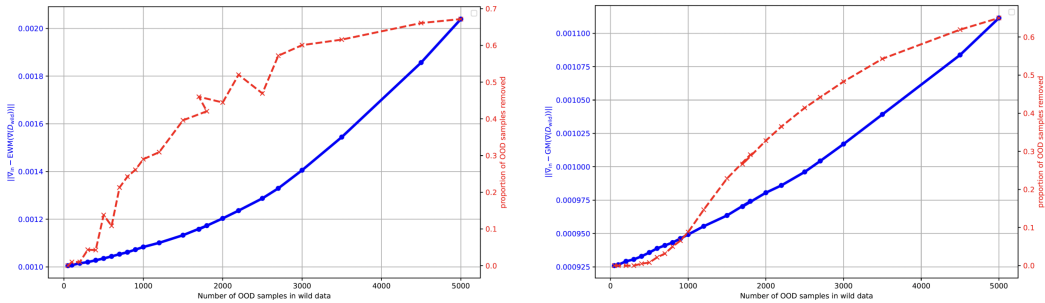


Figure 3: Comparison of element-wise median (EWM) and geometric median (GM) as the OOD proportion in the wild set varies.

Table 4: OOD detection performance of Medix and recent competitive baselines with CIFAR-100 as InD data, spanning a diverse suite of OOD datasets. Results are averaged over five runs; the best results in each column are highlighted in **bold**.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

dataset and train a ResNet-34 model¹ following the setup of Ming et al. (2023); Zhang et al. (2024). The model is trained with stochastic gradient descent using momentum 0.9 and weight decay 10^{-4} for 500 epochs, with an initial learning rate of 0.5 under cosine scheduling and a batch size of 512. We use the checkpoints provided by Ming et al. (2023)². As shown in Table 4, Medix outperforms even these competitive baselines by a significant margin—including those trained with contrastive objectives—achieving an average FPR95 of 8.68% and an average AUROC of 97.76%, and thereby demonstrating that its advantages hold well beyond any single class of methods. In addition, Medix provides a provably low error rate, offering theoretical guarantees that most alternative approaches lack.

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

We next investigate whether Medix scales to large-scale wild OOD data under the more demanding unseen-OOD setting. This setting is challenging for two reasons: (1) it relies on a large-scale wild OOD dataset ($\mathbb{P}_{\text{out}}$), increasing both data complexity and computational demand, and (2) the test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) is deliberately chosen to differ distributionally from the wild OOD data used during training, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$, which more faithfully reflects the diverse and unpredictable deployment scenarios encountered in practice. We used CIFAR-100 as the labeled InD data, the 300K Random Images dataset Hendrycks et al. (2019) as the unlabeled wild OOD data, and SVHN as the test OOD data. This setup is especially demanding because the large-scale wild OOD data (300K Random Images) differs markedly from the OOD test data (SVHN), stressing the method’s ability to generalize beyond the wild distribution it was trained on. Comparing against baselines that likewise leverage wild data during training, the results in Table 5 highlight the superior performance of Medix, with a substantially lower FPR95 (41.29 ± 1.2) and higher AUROC (87.25 ± 0.6), confirming its effectiveness at distinguishing InD from OOD data even under pronounced distribution shift.

Table 5: OOD detection performance on large-scale, unseen OOD data, a challenging and realistic deployment regime.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

¹Since ResNet-34 is the standard model across these works, we adopt it to ensure consistency and a fair comparison.

²<https://github.com>

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

Another important question is whether the quality of the pseudo-labels $\hat{y}_{\tilde{x}_i}$, particularly their softmax confidence, affects OOD filtering, and whether a simple pre-filtering step that discards low-confidence pseudo-labels could improve robustness. To probe this, we evaluated how filtering low-confidence pseudo-labels affects OOD filtering, using CIFAR-10 as the InD dataset and LSUN-Resize as the OOD dataset as a representative case. Specifically, we computed the softmax probabilities for each pseudo-labeled sample and discarded low-confidence predictions using a threshold of 0.6, which removed 15.98% of the samples from the training set.

The outcome showed virtually no difference between the filtered and unfiltered variants: both yielded an FPR95 of 0.01%, and filtering raised AUROC only marginally, from 99.98% to 99.99%. Removing low-confidence pseudo-labels therefore has little impact on robustness or OOD detection, indicating that our method is broadly resilient to noisy or low-confidence labels and that additional filtering steps are unlikely to yield meaningful gains.

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We now turn to whether the Medix filtering phase is computationally feasible and memory-efficient at scale. To this end, we ran profiling experiments on an NVIDIA A100-SXM4-80GB GPU using roughly 15,000 unlabeled samples for each InD–OOD pair, with CIFAR-10 and CIFAR-100 as the InD datasets and LSUN-Resize as the OOD dataset. As Table 6 shows, GPU memory usage stays modest and the filtering phase completes within a tractable timeframe even when processing thousands of unlabeled samples. This indicates that Medix filtering is computationally feasible and imposes little memory overhead, and that it should scale gracefully to large datasets in general.

Table 6: Profiling results for Medix filtering on an NVIDIA A100 80GB GPU, indicating modest and scalable resource usage.

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Our work differs from recent semi-supervised open-set recognition methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) in several fundamental ways. For instance, in OpenML Saito et al. (2021) the central problem is handling outliers in semi-supervised learning (SSL) while training a standard classifier, whereas our central challenge is to detect OOD samples from unlabeled wild data and train a dedicated OOD detector. Put differently, whereas SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) aim to improve classification despite outliers, our approach targets OOD detection itself in the broad open-world setting where labeled OOD data is unavailable.

Second, these SSL methods seek a classifier that is robust to OOD samples, treating outliers as noise, whereas our goal is to explicitly detect OOD samples and train an OOD detector. In other words, SSL methods suppress OOD samples during training to improve SSL performance, while Medix actively detects and leverages those OOD samples to build a more effective and reliable OOD detection system.

Finally, the techniques differ: OpenMatch is based on consistency regularization, enforcing agreement across stochastic transformations of the same input via soft constraints on the classifier’s outputs that encourage smooth decision boundaries. Medix, by contrast, relies on median-based filtering for outlier detection, exploiting the statistical robustness of the median—its insensitivity to extreme values—to surface potential OOD samples. Its theoretical analysis derives error bounds from the contamination effect (the proportion of OOD samples) and the concentration effect (the sub-Gaussian behavior of InD gradients), and these general ingredients keep the OOD detection error low across settings.

B Broader Impacts and Limitations

As machine learning continues to advance, addressing OOD detection has become essential for robust and reliable model performance across the many domains in which such systems operate. We introduce a general median-centric framework for OOD detection that improves OOD handling and model safety. The impact of this research extends well beyond theoretical advances: because the framework is model- and domain-agnostic, it is naturally applicable to high-stakes areas such as healthcare, autonomous systems, and finance, among others. By strengthening OOD detection, we address a key obstacle to trustworthy model deployment and help foster broader adoption of machine learning technologies. Future work could explore integrating Medix with generative models for outlier synthesis or adapting it to dynamic environments where OOD distributions evolve over time, further widening its range of applicability.

A potential limitation of Medix is its reliance on a mixture of unlabeled InD and OOD data, which in practice may be noisy, corrupted, or inconsistently sampled. This assumption, however, is not specific to our method—it reflects the inherent uncertainty of real-world deployment environments and is a common starting point in robust learning theory, so it applies broadly rather than narrowly. To keep our assumptions explicit and verifiable, we formalize them rigorously in the main theorems and give a precise geometric condition in Remark 4.3 justifying the separation needed for reliable filtering. We further extend the guarantees to looser settings in Theorem C.3, removing the sub-Gaussian tail requirement and demonstrating resilience under substantially weaker distributional assumptions. These efforts underscore that, while Medix does make simplifying assumptions, they are both interpretable and relaxable, providing a principled and broadly applicable foundation for future advances in OOD detection under realistic noise conditions.

785 C Proof of Theorem

786 For clarity and consistency across the theoretical and empirical sections, Table 7 summarizes the key
 787 notation used throughout the paper, covering gradient-related quantities, sample sizes, filtering terms,
 788 and the parameters appearing in our concentration bounds. Unless stated otherwise, we use boldface
 789 for vectors, write $\mathcal{X}$ for the (generic) input space, and assume all gradients are computed with respect
 790 to a fixed pretrained model f_ϕ .

Table 7: Summary of the notation used throughout the paper. We use bold symbols for vectors and $\mathcal{X}$ to denote the generic input space.

Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors

791 C.1 Bound on the Inlier Misclassification Rate

792 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
 793 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, and*
 794 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
 795 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

796 *Proof.* We structure the argument into two key steps that together apply to any InD gradient distribu-
 797 tion satisfying the stated assumption. In **Step 1**, we establish a high-probability concentration bound
 798 for the gradients of InD samples around their mean $\bar{\nabla}_{\text{in}}$, and we call an InD sample whose gradient
 799 lies within this high-probability region a *good inlier*. In **Step 2**, we use a “swapping” argument on the
 800 EWM objective to show that removing more than a $(2\eta + \frac{\pi}{2}(1 - \pi))$ fraction of *good inliers*—where
 801 η is the fraction of InD samples with gradients outside the high-probability region (the non-good

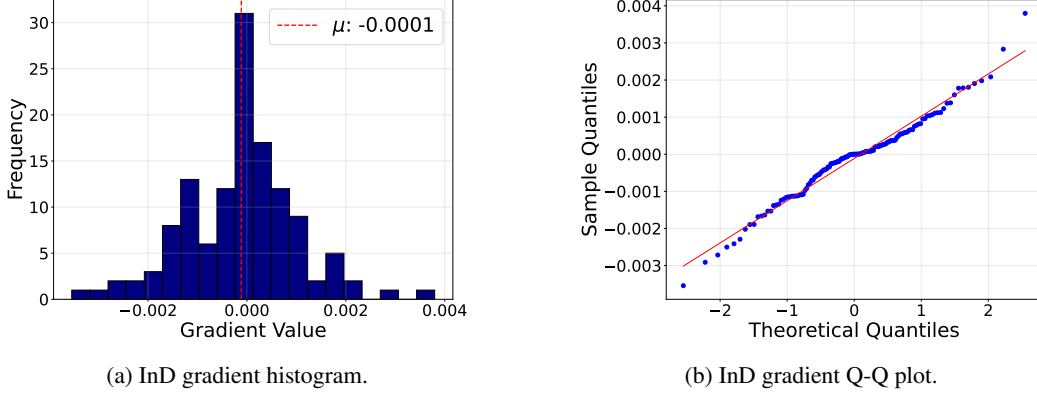


Figure 4: Empirical evidence that InD sample gradients exhibit coordinate-wise sub-Gaussian behavior, supporting the generality of our assumption. (left) Histogram of gradient values (CIFAR-100 InD data) showing concentration around the mean with light tails, consistent with sub-Gaussianity. (right) Q-Q plot comparing empirical quantiles of InD gradients against a theoretical Gaussian, confirming alignment with sub-Gaussianity.

inliers)—would contradict optimality. Combining these two steps yields the stated upper bound on the inlier misclassification rate ERR_{in} .

Step 1: Concentration of InD Gradients. Let $\nabla \ell(f_\phi(\mathbf{x})) \in \mathbb{R}^d$ be the gradient (with respect to ϕ) of a loss evaluated at an InD sample $\mathbf{x}$. Suppose each coordinate of this gradient is sub-Gaussian with variance proxy σ^2 , centered at the same mean vector $\bar{\nabla}_{\text{in}}$. That is, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla \ell(f_\phi(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

Define

$$\mathbf{v} := \nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

It is straightforward to show that

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_\infty,$$

so if $\|\mathbf{v}\|_2 > \epsilon \sqrt{d}$ then there must exist at least one coordinate j with $|v_j| > \epsilon$.

Hence, combining (6) across $j = 1, \dots, d$ via a union bound yields

$$\mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the total number of InD points in the set $\mathcal{S}_{\text{wild}}$. We define an InD point $\mathbf{x}$ to be *bad* if

$$\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}. \quad (8)$$

Our goal is to show that, with high probability, only a small fraction of these m_{in} inliers can be bad.

From our sub-Gaussian assumption and applying the union bound across the d coordinates, we define a constant $p \in [0, 1]$ as follows:

$$p := \mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

Here, $\mathbf{x}$ is a generic InD sample, and we assume that the samples are i.i.d. across the m_{in} inliers.

Let X be the random variable denoting the number of bad inliers among the m_{in} samples. For each inlier $\mathbf{x}_i$, define the indicator variable

$$I_i := \begin{cases} 1, & \text{if } \|\nabla \ell(f_\phi(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

819 Then, X can be expressed as

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

820 By the linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

821 Observing that X is a sum of m_{in} Bernoulli(p) random variables, each bounded between 0 and 1, we
822 can apply Hoeffding's inequality to obtain, for any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

823 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log(\frac{1}{\delta})}{2}}, \quad (17)$$

824 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

825 With probability at least $1 - \delta$, we therefore have

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

826 Dividing both sides by m_{in} yields an upper bound on the fraction of bad inliers:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

827 Let us denote this fraction by η . By setting $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$ Thus, with probability at least
828 $1 - \delta$, no more than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients lying
829 outside the radius $\epsilon\sqrt{d}$ around $\bar{\nabla}_{\text{in}}$. These are precisely the *bad inliers*, while the remainder are
830 called *good inliers*.

831 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 832 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

833 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 834 thereby contradicting the optimality of $\mathcal{S}^*$.

835 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 836 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

837 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

838 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

839 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (26)$$

840 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 841 available for swapping in.

842 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 843 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

844 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

845 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

846 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

847 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

848 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 849 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

850 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

851 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (33)$$

852 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (35)$$

853 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (36)$$

854 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

855 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
856 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

857 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
858 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
859 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

860 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

861 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
862 increase it, so the median distance from μ_j does not increase and possibly decreases.

863 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

864 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

865 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
866 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

867 □

868 C.2 Bound on the Outlier Retention Rate

869 We now establish a non-asymptotic upper bound on the fraction of OOD points that remain in
870 the subset returned by the MEDIX filter. This result mirrors the inlier misclassification bound
871 (Appendix C.1), but rather than relying on inlier concentration it exploits a general separation
872 assumption between the OOD and InD gradient means, and it holds under mild conditions.

873 **Theorem C.2** (Outlier Retention Bound under Vector Separation). Assume that for every OOD point
874 $x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:

875 **(I) Sub-Gaussianity.** Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .

876 **(II) Vector Mean Separation.** The mean OOD gradient vector μ_{out} satisfies

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d}, \quad \text{for some } \Delta > 0.$$

877 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

878 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

879 *Proof. Step 1: Concentration of OOD Gradients.*

880 By assumption, each coordinate of $g(x)$ is sub-Gaussian, and the mean vector is separated from $\bar{\nabla}_{\text{in}}$
881 by at least $\Delta\sqrt{d}$. Therefore, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

882 has probability at most $p_{\text{out}}(\varepsilon)$.

883 **Counting the “ambiguous” OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

884 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

885 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding’s inequality states that for any $s > 0$

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

886 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

887 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

888 which holds with probability at least $1 - \delta$.

889 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

890 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_\varepsilon^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_\varepsilon^* > \frac{1-\pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

Thus, more than half the points in $\mathcal{S}_{\text{in}}^*$ are separated OODs whose gradients lie outside the ball

$$B_\varepsilon := \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon \sqrt{d} \right\}. \quad (46)$$

By the definition of the coordinate-wise median, this implies that in every coordinate, the majority of entries in $G_{\mathcal{S}_{\text{in}}^*}^{(j)}$ lie outside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, hence:

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}^*}) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

Now consider a new subset $\hat{\mathcal{S}}$ formed by swapping the m_{sep}^* separated OODs in $\mathcal{S}_{\text{in}}^*$ with an equal number of InD points whose gradients lie inside B_ε . Such inliers exist because fewer than $m_{\text{out}} \eta_{\text{out}}$ are ambiguous, and the total number of InD points is m_{in} .

Then at least half the entries in every coordinate of $G_{\hat{\mathcal{S}}}^{(j)}$ lie inside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, implying:

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

Combining with (47) yields a contradiction:

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}^*})\|_2^2. \quad (49)$$

This contradicts the optimality of $\mathcal{S}_{\text{in}}^*$, and so the assumption must be false. Therefore, our assumption must be false. With probability at least $1 - \delta$, we conclude that the outlier retention rate is bounded by

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1-\pi}{2\pi}, \quad (50)$$

□

C.3 Theorem C.3 without Sub-Gaussianity

Theorem C.3 (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^2 = \sigma^2, \quad \mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^4 \leq \mu_4.$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{2 \left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

911 **Differences with the original bound.** The original bound, which assumes sub-Gaussianity, features
 912 a decay term

$$2 \exp\left(-\frac{\epsilon^2 d}{2\sigma_{\text{sub}}^2}\right)$$

913 that decreases exponentially with dimension d , making the bound extremely tight for large d . By
 914 contrast, the broader bound here relies only on finite fourth moments and exhibits a decay term

$$\frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}$$

915 that decays as $O(1/d)$, yielding a looser guarantee for large d . The trade-off is deliberate: whereas
 916 the original bound requires sub-Gaussian tails, this version only assumes finite fourth moments and
 917 thus accommodates a wider class of distributions with heavier tails. Notably, both bounds share the
 918 finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

919 which shrinks as m_{in} grows, independent of the tail assumptions.

920 *Proof.* The proof follows the structure of the original Theorem 4.1 and again consists of two main
 921 steps: (1) establishing a concentration bound on the fraction of “bad” InD samples (those far from
 922 ∇_{in}) without relying on sub-Gaussianity, and (2) using a swapping argument to bound the inlier
 923 misclassification rate. The only change is that the sub-Gaussian tail bound is replaced by a more
 924 broadly applicable moment-based bound derived from the finite fourth-moment assumption.

925 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

926 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
 927 and we assume ∇_{in} is the population mean. An InD sample is classified as “bad” if:

$$\|v\|_2 > \epsilon \sqrt{d}, \quad (52)$$

928 and “good” otherwise. Our goal is to bound the fraction of bad inliers among the m_{in} InD samples in
 929 S_{wild} .

930 Since $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, we analyze the ℓ_2 norm via the sum of squared coordinates. Assume that
 931 the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$, each with:

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

932 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E}\left[\sum_{j=1}^d v_j^2\right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var}\left(\sum_{j=1}^d v_j^2\right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

933 Compute the variance of v_j^2 :

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

934 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

935 Using Chebyshev's inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

936 We need $\mathbb{P}(\|v\|_2 > \epsilon\sqrt{d})$, which corresponds to $\|v\|_2^2 > \epsilon^2 d$. Set the threshold:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

937 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

938 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

939 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
940 bad with probability at most p . Then:

$$\mathbb{E}[X] \leq m_{\text{in}}p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

941 Since $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ if the i -th InD sample is bad and 0 otherwise, and I_i are i.i.d.
942 Bernoulli(p) variables bounded in $[0, 1]$, apply Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

943 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

944 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

945 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

946 This η represents the upper bound on the fraction of InD samples whose gradients deviate from $\bar{\nabla}_{\text{in}}$
 947 by more than $\epsilon\sqrt{d}$, relying only on finite fourth moments and coordinate independence.

948 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 949 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (68)$$

950 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 951 thereby contradicting the optimality of $\mathcal{S}^*$.

952 Let m_ϵ be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 953 lies outside the closed ℓ_2 ball

$$B_\epsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \epsilon\sqrt{d} \right\}. \quad (69)$$

954 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

955 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

956 Define the swap size as

$$m'_\epsilon := \min \{ m_\epsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}} \}. \quad (71)$$

957 This quantity counts how many good inliers (those inside B_ϵ) are currently excluded from $\mathcal{S}^*$ and
 958 available for swapping in.

959 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ϵ points from $\mathcal{S}^*$ that lie outside B_ϵ (OOD points or bad inliers)
 960 and inserting m'_ϵ good inliers from B_ϵ in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

961 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\epsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

962 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\epsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

963 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

964 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

965 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 966 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

967 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

968 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta) m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

969 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

970 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

971 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

972 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
 973 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

974 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
 975 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
 976 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

977 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

978 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
 979 increase it, so the median distance from μ_j does not increase and possibly decreases.

980 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

981 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

982 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
 983 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

984 □